

II. $(10x + y) - (10y + x) = 9 \Rightarrow x - y = 1.$

III. $x - y = 1.$

Thus, I and II as well as I and III give the answer.

$\therefore$ Correct answer is (e).

- 17. I.** Let the ten's and unit's digit be $3x$ and $2x$ respectively.

II. $(30x + 2x) - (20x + 3x) = 18 \Leftrightarrow x = 2.$

III. $3x \times 2x = 24 \Leftrightarrow x^2 = 4 \Leftrightarrow x = 2.$

Thus, any two of the three will give the answer.

$\therefore$ Correct answer is (a).

- 18.** Let the required numbers be x and y , where $x > y$.

I. $x - y = 6$... (i)

II. $\frac{30}{100}x = \frac{40}{100}y \Leftrightarrow 3x - 4y = 0$... (ii)

III. $\frac{\frac{1}{2}x}{\frac{1}{3}y} = \frac{2}{1} \Leftrightarrow \frac{3x}{2y} = \frac{2}{1} \Leftrightarrow \frac{x}{y} = \frac{4}{3} \Leftrightarrow 3x - 4y = 0$... (iii)

So, we may solve (i) and (ii) or (i) and (iii) together to find x and y .

Thus, I, and either II or III together give the answer.

$\therefore$ Correct answer is (e).

- 19.** Let the ten's and unit's digit be x and y respectively.

I. $(10x + y) - (10y + x) = 27 \Leftrightarrow x - y = 3.$

II. $x - y = 3.$

III. $x - y = 3.$

Thus, even all the given three statements together do not give the answer.

$\therefore$ Correct answer is (e).

- 20.** Let the ten's digit be x and the unit's digit be y .

I. $x = y^3$... (i)

II. $x = 4y$... (ii)

III. $x \neq y$... (iii)

From (i) and (ii), we have : $y^3 = 4y \Leftrightarrow y^3 - 4y = 0$
 $\Leftrightarrow y(y^2 - 4) = 0$

$\Leftrightarrow y^2 - 4 = 0$ [$\because y \neq 0$]

$\Leftrightarrow y^2 = 4 \Leftrightarrow y = 2.$

So, $x = y^3 = 2^3 = 8.$

$\therefore$ The required number is 82.

Thus, I and II together give the answer.

$\therefore$ Correct answer is (a).

- 21.** Let the three consecutive even numbers be x , $(x + 2)$ and $(x + 4)$.

I. $\frac{(x + 4) + (x + 6) + (x + 8) + (x + 10)}{4} = 17$

$\Leftrightarrow 4x + 28 = 68 \Leftrightarrow 4x = 40 \Leftrightarrow x = 10.$

So, the required numbers are 10, 12 and 14.

II. $(x + 4) - x = 4 \Leftrightarrow 4 = 4.$

So, the value of x cannot be determined.

III. $x^2 + (x + 2)^2 + (x + 4)^2 = 440$

$\Leftrightarrow x^2 + x^2 + 4 + 4x + x^2 + 16 + 8x = 440$

$\Leftrightarrow 3x^2 + 12x - 420 = 0$

$\Leftrightarrow x^2 + 4x - 140 = 0$

$\Leftrightarrow x^2 + 14x - 10x - 140$

$\Leftrightarrow x(x + 14) - 10(x + 14) = 0$

$\Leftrightarrow (x + 14)(x - 10) = 0$

$\Leftrightarrow x = 10.$

So, the required numbers are 10, 12 and 14.

Thus, I alone or III alone gives the answer.

$\therefore$ Correct answer is (d).

- 22.** Let the ten's digit be x and unit's digit be y .

(I and II). $x + y = 14$ or $y = (14 - x)$... (i)

And, $[10(14 - x) + x] - [10x + (14 - x)]$

$= 18 \Leftrightarrow (140 - 9x) - (9x + 14) = 18$

$\Leftrightarrow 18x = 108 \Leftrightarrow x = 6.$

So, $y = 14 - 6 = 8.$

$\therefore$ Required number = 68.

(II and III). $x + y = 14$... (ii)

and $x - y = \pm 2$... (iii)

Solving (ii) and (iii), we get : $x = 6$ or $8.$

If $x = 6$, $y = 8$ & If $x = 8$, $y = 6.$

$\therefore$ Required number is either 68 or 86.

(I and III). Since the number obtained by interchanging the digits is greater, the ten's digit is smaller than the unit's digit.

$x + y = 14$... (iv)

And, $y - x = 2$... (v)

Solving (iv) and (v), we get : $y = 8.$

So, $x = 14 - y = 6.$

$\therefore$ Required number = 68.

Thus, II and either I or III gives the answer.

$\therefore$ Correct answer is (c).